

Generative AI and Labor Market Outcomes: Evidence from the United Kingdom

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Abstract

This paper examines the effects of LLMs on UK labor market outcomes. We use a difference-in-differences design that compares outcomes across firms and occupations from 2021 to 2025 based on their differential exposure to LLM capabilities. Our results show that highly exposed firms reduce employment, with effects concentrated in junior positions. These firms sharply curtail new hiring, with technical and creative roles experiencing the steepest declines. At the occupation level, roles more exposed to LLMs also show substantial reductions in job listings. The observed effects concentrate almost entirely in high-wage segments. Hence, our findings suggest that LLMs may compress wage inequality.

Keywords: Generative AI; Large Language Models; Automation; Labor demand; United Kingdom.

JEL: O33; J23; J31

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1 Introduction

The release of ChatGPT in November 2022 marked the beginning of large-scale diffusion of large language models (LLMs) around the world. Within months, millions of workers gained access to tools capable of drafting documents, writing code, analyzing data, and performing countless other cognitive tasks that previously required human expertise. This rapid integration into workplace workflows has generated widespread anxiety about the future of work, technological unemployment, and economic disruption, with some early evidence showing large productivity gains (Brynjolfsson, Li and Raymond, 2025; Noy and Zhang, 2023).

This uncertainty reflects a potential shift in the nature of this technological innovation. Unlike previous technologies that predominantly displaced routine tasks in lower- and middle-skilled occupations (Autor, 2015; Autor et al., 2003), LLMs possess capabilities that extend across the entire skill distribution, with the potential to affect highly skilled professional and analytical roles as well (Eloundou et al., 2024). Estimates suggest that around 80% of the U.S. workforce could see at least 10% of their tasks affected, with exposure rising in higher-income occupations on average (Eloundou et al., 2024). As such, the introduction of generative AI raises important questions about whether LLMs will erode job security, depress white-collar wages, and increase or decrease inequality.

To examine these questions, we use detailed UK labor market data from Revelio Labs, covering millions of firms and job postings between 2021 and 2025. Our identification strategy exploits the sudden and arguably exogenous introduction of ChatGPT in November 2022. The impact of this shock varied across firms and industries depending on their occupational mix and the extent to which core activities are susceptible to LLM automation. We measure this heterogeneity using the occupational level LLM exposure scores developed by Eloundou et al. (2024), who evaluate 19,265 detailed work activities from the O*NET database using both human annotators and GPT-4 to determine whether LLM assistance could reduce task completion time by at least 50% while maintaining quality. These exposure scores allow us to quantify the share of tasks within each occupation, or the share of jobs within each firm that can be performed by LLMs.¹

¹Their measure strongly correlates with other LLM exposure measures (see e.g. Tomlinson et al., 2025; Hampole et al., 2025).

Building on these exposure measures, we estimate a series of difference-in-differences models around the introduction of ChatGPT that exploit firm- and occupation-level variation in LLM exposure to estimate the causal effects of the ChatGPT shock on employment, wages, pay dispersion, job seniority, workplace arrangements, and overall hiring dynamics.

Our results show a clear impact of LLM introduction on UK labor markets, with effects emerging gradually across multiple dimensions of employment. At the firm level, a one standard deviation increase in LLM exposure is associated with a 0.3 percent reduction in total employment. The impact is concentrated among junior positions, with low-seniority employment declining by 0.4 percent while high-seniority positions show smaller and partly insignificant reductions. This compositional shift reduces the fraction of low-seniority workers, which mechanically raises average compensation in exposed companies as the workforce skews toward higher-paid senior employees.

These patterns also show up in hiring intentions. In an analysis of firm-level job posting data, we find that a one standard deviation increase in LLM exposure reduces the likelihood that firms post job vacancies in a given month by 1.0 percentage point. The reduction in job postings among exposed firms is strongest for technical and creative roles such as software engineers and designers. At the occupation level, we observe even more pronounced effects on hiring. A one standard deviation increase in an occupation’s LLM exposure is associated with a 3.9 percent decline in job postings, accompanied by substantial reductions in posted salaries.

Importantly, these effects are not uniformly distributed across the labor market. Our heterogeneity analysis shows that LLM disruption is concentrated almost entirely in higher-wage segments. High-compensation firms experience employment reductions of 0.7 percent per standard deviation increase in exposure, compared with near-zero effects for low-compensation firms. Similarly, high-salary occupations experience 5.7 percent declines in job postings per standard deviation, while low-salary roles show no significant change. These results point in the opposite direction from historical automation experiences, in which routine middle-skill occupations bore the brunt of technological displacement (Autor, 2024).

To interpret these results, we draw on the task-based framework (Autor et al., 2003; Acemoglu and Restrepo, 2018, 2019, 2020; Autor and Thompson, 2025). This framework models production as discrete tasks allocated between labor and capi-

tal based on comparative advantage. When automation enables capital to perform tasks previously done by labor more cheaply, it creates a *displacement effect* that reduces labor demand. Conversely, the creation of new tasks where labor maintains comparative advantage generates a *reinstatement effect* that increases labor demand (Acemoglu and Restrepo, 2019; Restrepo, 2024). For LLMs, which can automate cognitive tasks previously considered non-routine, the framework predicts that displacement will dominate in the short run, particularly for entry-level positions involving routine information processing and basic analysis. Senior roles requiring tacit knowledge, judgment, and complex interpersonal coordination should prove less vulnerable. This is consistent with our findings.

Our findings contribute to the emerging international evidence on labor market impacts of AI (del Rio-Chanona et al., 2025). In the United States, Brynjolfsson, Chandar and Chen (2025); Lichtinger and Hosseini Maasoum (2025); Dominski and Lee (2025) document employment declines in exposed industries concentrated in early-career positions. However, Chandar (2025) finds no employment effects using CPS data, and Humlum and Vestergaard (2025) report null results from Danish administrative data. These disparities may reflect several factors: differences in labor market institutions between countries, variation in the timing of LLM adoption across regions, and methodological choices regarding exposure measurement and outcome variables. Our UK evidence suggests that effects manifest differently across adjustment margins, with hiring responding more quickly and strongly than employment stocks.

Methodologically, our paper builds on recent work that also uses Revelio Labs data to construct firm-level AI exposure measures. Eisfeldt et al. (2023) show that higher-exposure US firms experience excess stock returns following the release of ChatGPT, while Hampole et al. (2025) develop task-level exposure measures varying across firms and occupations to study US labor demand between 2010 and 2023. We extend their work in three directions: we shift focus to the UK labor market, examine realized employment and job posting outcomes (as opposed to Eisfeldt et al. (2023)’s focus on financial data), and exploit data through mid-2025 to capture labor market adjustment to generative AI adoption (as opposed to Hampole et al. (2025)’s focus on earlier waves of AI).

In the UK context, we extend Henseke et al. (2025), who develop a Generative AI Susceptibility Index and provide suggestive evidence that wage premia decline